

3. Given the function $h(x) = x + 3$ is a factor of $f(x) = x^4 - x^3 + kx^2 + 14x - 12$. Then find the constant k and calculate all roots of $f(x)$. (5.5 point)

Soln

from the remainder theorem, if $(x+a)$ is factor of $f(x)$, then $f(-a) = 0$.

$$\therefore f(-3) = (-3)^4 - (-3)^3 + k(-3)^2 + 14(-3) - 12 = 0$$

$$\Rightarrow 81 + 27 + 9k - 42 - 12 = 0$$

$$9k + 54 = 0$$

$$9k = -54$$

$$k = -6$$

The equation will be

$$f(x) = x^4 - x^3 - 6x^2 + 14x - 12$$

To find the roots,

~~f~~ = we know that $x+3$ is factor of $f(x)$

by using division, the equation will be

$$(x+3)(x^3 - 4x^2 + 6x - 4) = 0$$

now $x = -3$ is one root of $f(x)$.

$$x^3 - 4x^2 + 6x - 4 = 0$$

from here we have

$$P = \pm 1, \pm 2, \pm 4$$

$$f(1) = 1 - 4 + 6 - 4 = -1 \neq 0, \quad 1 \text{ is not the root of } f(x).$$

$$f(2) = 8 - 16 + 12 - 4 = 0 \Rightarrow 2 \text{ is the root of } f(x)$$

Since $f(2)$ is the root of $f(x)$, then

$$(x-2) \text{ is factor of } x^3 - 4x^2 + 6x - 4$$

$\therefore$ by using the long division the equation will be

$$(x-2)(x^2 - 2x + 2)$$

from here we have $x=2$ is another root of $f(x)$.

$\rightarrow$ To find another roots of $f(x)$

$$x^2 - 2x + 2 = 0$$

by using the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad a=1, \quad b=-2, \quad c=2$$

$$x = \frac{2 \pm \sqrt{4-8}}{2} = \frac{2 \pm \sqrt{-4}}{2}$$

$$x = \frac{2 \pm 2i}{2} \quad x = \frac{1 \pm i}{1}$$

$$x = 1-i, \quad x = 1+i$$

The roots of $x^4 - x^3 - 6x^2 + 14x - 12$ are

$$x = -3$$

$$x = 2$$

$$x = 1+i$$

$$x = 1-i$$

4. Find the domain, intercepts, asymptotes, and sketch the graph of the function

$$f(x) = \frac{x^3 + 6x^2 + 5x}{x^2 - 3x - 4}$$

$$f(x) = \frac{x^3 + 6x^2 + 5x}{x^2 - 3x - 4}$$

(6.5 point)

a) To find domain, $x^2 - 3x - 4 = 0$

$$(x-4)(x+1) = 0$$

$$x = 4, x = -1$$

$$\text{Dom}(f) = \{x \mid x \neq -1, 4\}$$

To simplify the given equation

$$\frac{x(x+5x+1)}{(x-4)(x+1)} = \frac{x(x+1)(x+5)}{(x-4)(x+1)} = \frac{x(x+5)}{x-4}, \quad x \neq -1$$

b) To find intercepts

i) x-intercept, at $y = 0$

$$0 = \frac{x^3 + 6x^2 + 5x}{x^2 - 3x - 4} = \frac{x(x+5)}{x-4}$$

$$x(x+5) = 0, \quad x = 0, \text{ or } x = -5$$

$$(0, 0), (-5, 0)$$

ii) y-intercept, at $x = 0$

$$y = \frac{0(0+5)}{0-4} = 0, \quad (0, 0)$$

c) V. Asymptote, as $x \rightarrow 4^+$, $f(x) \rightarrow +\infty$

$$x \rightarrow 4^-, f(x) \rightarrow -\infty$$

Vertical asymptote at $x = 4$

i) The graph has no horizontal asymptote

ii) Oblique asymptote, by using long division

$$\begin{array}{r} x+9 \\ x-4 \overline{) x^2+5x} \\ \underline{x-4} \\ 9x+36 \\ \underline{9x-36} \\ 72 \end{array}$$

$$\begin{array}{r} x+9 \\ x-4 \overline{) x^2+5x} \\ \underline{x-4} \\ 9x+36 \\ \underline{9x-36} \\ 72 \end{array}$$

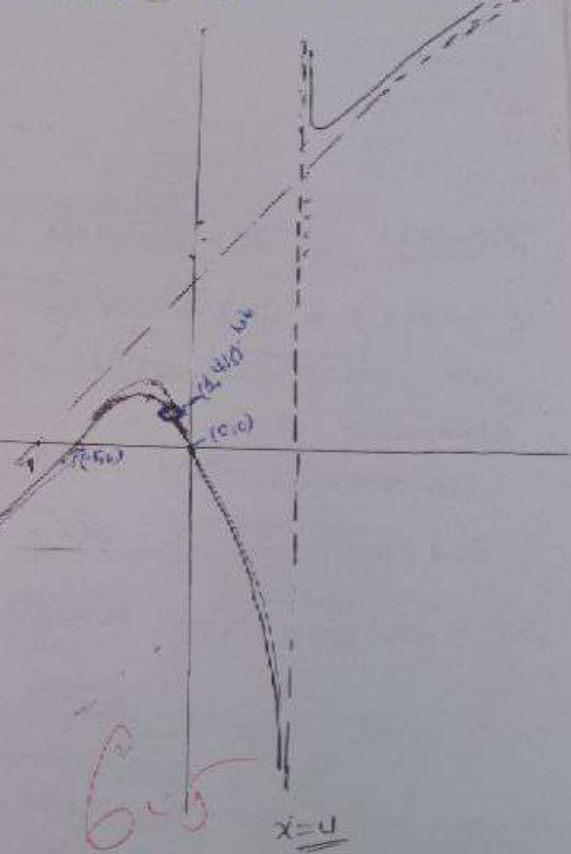
$\therefore y = x+9$ is oblique asymptote

$$\text{as } x \rightarrow 4^+, f(x) \rightarrow +\infty$$

$$\text{as } x \rightarrow 4^-, f(x) \rightarrow -\infty$$

$$\text{as } x \rightarrow \infty, f(x) = x+9$$

$\therefore y = x+9$ is oblique asymptote



2. Solve the equation

$$(z^4 - 17)^{1/6} = i$$

(6 points)

When we simplify the equation

$$(z^4 - 17)^{1/6} = i$$

$$z^4 - 17 = -1 \quad i^6 = -1$$

$$z^4 = 16$$

$$z = (16 + 0i)^{1/4}$$

$$r = \sqrt{16^2 + 0^2} = 16$$

$$\theta = \tan^{-1}\left(\frac{0}{16}\right) = 0$$

now by using $C_k = r^{1/n} e^{i\left(\frac{\theta}{n} + \frac{2k\pi}{n}\right)}$, $n=4$
 $k=0,1,2,3$

$$C_0 = 16^{1/4} e^{i(0)}$$

$$= 2[\cos 0 + i \sin(0)]$$

$$= 2[1 + 0i] = \underline{2}$$

$$C_1 = 16^{1/4} e^{i\left(0 + \frac{2\pi}{4}\right)} = 2e^{i(\pi/2)}$$

$$C_1 = 2[\cos \pi/2 + i \sin \pi/2]$$

$$= 0 + 2i = \underline{2i}$$

$$C_2 = 16^{1/4} e^{i(0 + \pi)} = 2e^{i\pi}$$

$$C_2 = 2[\cos \pi + i \sin \pi]$$

$$= -2 + 0i = \underline{-2}$$

$$\therefore C_3 = 16^{1/4} e^{i\left(0 + \frac{6\pi}{4}\right)} = 2e^{i\left(\frac{3\pi}{2}\right)}$$

$$C_3 = 2\left[\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right]$$

$$= 2[0 - i] = \underline{-2i}$$

Therefore the solution set
of $(z^4 - 17)^{1/6} = i$ are
 $ss = \{-2, 2i, 2, 2i\}$

Part-II: Workout problems

Show all the necessary steps clearly and neatly on the space provided.

1. Using the truth table investigate the validity of the following argument.

Whenever he participates a race, he is the winner. Either he participates the race or he is sick. He is not the winner. Therefore, he is sick.

(6 point)

let p : he participates a race
 q : he is the winner
 r : he is sick.

The argument is:

$p \Rightarrow q, p \vee r, \neg q \vdash r$

In 7th row our premises are true but the same time the conclusion is also true.

p	q	r	$p \Rightarrow q$	$p \vee r$	$\neg q$	r
T	T	T	T	T	F	T
T	T	F	F	T	F	F
T	F	T	F	T	T	T
T	F	F	F	T	T	F
F	T	T	T	T	F	T
F	T	F	T	F	F	F
F	F	T	T	T	T	T
F	F	F	T	F	T	F

In 7th row our conclusion is true and at the same time all premises are true. therefore the argument is valid.

5. Find the equation of the hyperbola, center, foci, vertices, and asymptotes, and sketch the hyperbola satisfying the condition one focus at $(-3 - 3\sqrt{13}, 1)$, asymptotes intersecting at $(-3, 1)$, and one asymptote passing through the point $(1, 7)$. (6.5 point)

~~Given data~~ Soln
 $F(-3, -3\sqrt{13}, 1)$, intersection of asympt $(-3, 1)$
 $P(1, 7)$

We know that the asymptote of hyperbola intersect each other at the center of the hyperbola.
 So $C(-3, 1)$ is center

Since y component of the focus and the center is the same, the given hyperbola is horizontal hyperbola.
 Its equation is given by

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

To find the ~~foci~~ focus, we have to know the 'c'.
 c is the distance b/w focus and the center

$$c = \sqrt{(-3 - 3\sqrt{13} - (-3))^2 + (1 - 1)^2} = \sqrt{(3\sqrt{13})^2} = 3\sqrt{13}$$

$F = (h \pm c, k)$, $\therefore F(-3 \pm 3\sqrt{13}, 1)$
 The foci are $F(-3 + 3\sqrt{13}, 1)$, $F(-3 - 3\sqrt{13}, 1)$

from the definition of the hyperbola
 $|PF_1| - |PF_2| = 2a$, $P(1, 7)$

$$\sqrt{(1 - (-3 + 3\sqrt{13}))^2 + (7 - 1)^2} - \sqrt{(1 - (-3 - 3\sqrt{13}))^2 + (7 - 1)^2} = 2a$$

$$= \sqrt{169 - 12\sqrt{13}} - (\sqrt{169 + 12\sqrt{13}}) = 2a$$

$$(\sqrt{169 - 12\sqrt{13}})^2 = 2a^2 + \sqrt{169 - 12\sqrt{13}} \cdot \sqrt{169 + 12\sqrt{13}} + 169 + 12\sqrt{13}$$

$$169 - 12\sqrt{13} = 4a^2 + 169 + 12\sqrt{13}$$

By using another method
 $c = 3\sqrt{13}$, $c^2 = 117$
 $b^2 = 117 - a^2$, by substituting this in the original equation we get
 $\frac{(x+3)^2}{a^2} - \frac{(y-1)^2}{117-a^2} = 1$, $P(1, 7)$
 $\frac{(1+3)^2}{a^2} - \frac{(7-1)^2}{117-a^2} = 1$

Since the two asymptotes pass through $(-3, 1)$, $y - k = \pm \frac{b}{a}(x - h)$
 $(7 - 1) = \pm \frac{b}{a}(1 + 3)$, $a(-3, 1)$
 $6 = \pm \frac{b}{a}(4) \Rightarrow \frac{b}{a} = \frac{3}{2}$, $b = \frac{3a}{2}$

$$c^2 = a^2 + b^2$$

$$117 = a^2 + \frac{9a^2}{4}$$

$$4(117) = 4a^2 + 9a^2$$

$$468 = 13a^2$$

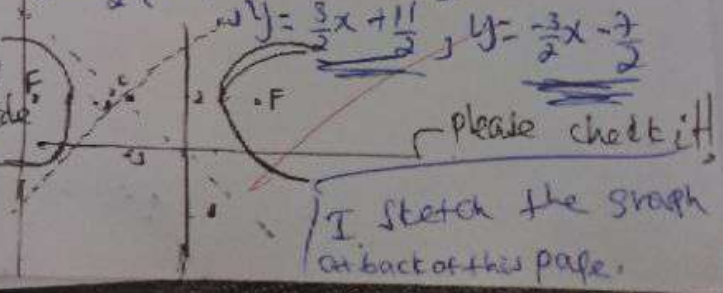
$$a^2 = 36, a = 6$$

$$b = \frac{3a}{2} = 9$$

The equation will be
 $\frac{(x+3)^2}{36} - \frac{(y-1)^2}{81} = 1$

The Vertices, $V(h \pm a, k) = V(-3 \pm 6, 1)$
 $V(-9, 1)$, $V(3, 1)$

The asymptotes be $y - k = \pm \frac{b}{a}(x - h)$
 $y - 1 = \pm \frac{3}{2}(x + 3)$
 $y = \frac{3}{2}x + \frac{11}{2}$, $y = -\frac{3}{2}x - \frac{7}{2}$



FINAL EXAMINATION COVER PAGE

Part-I: (Short Answer) Give the most simplified answers on the space provided. [Each blank space worths 1.5 points]

1. The relation $R = \{(x, y) : y \text{ is the mother of } x\}$ satisfies a function. (True/False)

TRUE

2. Assume $\neg p, q$ and $\neg r$ are false, then the truth value of $(\neg r \wedge p) \Rightarrow [(q \wedge \neg p) \Rightarrow r]$ is

TRUE

3. Given two sets A and B,

A. $(A' \cap B') - (A \cup B)' =$

B. If $B = \{1, 2, 3\}$ such that $A \cap B = \emptyset$ and $A \cup B = U$, then $A' - B' =$

4. The imaginary part of the complex number $(1 - i)^{64}$ is

5. The lub and glb of the set $A = \left\{1 - \frac{(-1)^n}{n}\right\}_{n=1}^{\infty}$ are and respectively.

6. Given the functions $f(x) = \frac{x+7}{x-3}$, $g(x) = \frac{1}{x+4}$, and $h(x) = \cosh x$

- A. Domain of $(g \circ f)(x)$ is

- B. Range of $h(x)$ is

7. The solution of the equation $\log x + \log(x - 3) = 1$ is

8. Equation of the line tangent to the circle with the center at $(-1, 1)$ and point of tangency at $(-1, 3)$ is

9. Given the equation of a parabola $y^2 = 7x$, then the coordinates of its focus is

10. Equation of the ellipse whose endpoints of the major axis are at $(3, -4)$ and $(3, 4)$ and passes through the origin is

$$\begin{aligned} 70 &= 7 \times 10 \\ 70 &= 7 \times 10 \\ 70 &= 7 \times 10 \end{aligned}$$

$$14.25$$

$$\begin{aligned} (7F \wedge T) &\Leftrightarrow [(F \wedge F)] \\ F &\Leftrightarrow [F \Rightarrow T] \\ F &\Leftrightarrow T \end{aligned}$$

$$\{1, 2, 3\}$$

$$-4, 1, 3$$

$$\log_{10}(x^2 - 3x) = \log_{10}^n$$

$$\begin{aligned} x^2 - 3x &= 10 \\ x^2 - 3x - 10 &= 0 \end{aligned}$$

$$x_0 \quad y_0$$

$$\begin{aligned} (x-5)(x+2) &= 0 \\ x &= 5, x = -2 \end{aligned}$$

$$\frac{y-y_0}{x-x_0} = -\left(\frac{x_0-h}{y_0-k}\right)$$

$$\frac{y^2}{16} + \frac{(x-5)^2}{9} = 1$$